

LINEAR AND ROBUST CONTROL SYSTEMS

ELEC 9731

4A - SISO Classical Control Review

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Slide 1

Topics

- 1** Classical Control Review:
One Degree of Freedom Architecture
- CLTF, IDTF, NTF, Sensitivity Function.
- 2** Introduction to Two degree of Freedom Architectures.

Notation

$P(s) = P$ = open loop transfer function (OLTF)

$G(s) = G$ = closed loop transfer function (CLTF)

AIMS OF CONTROL

- Steer system outputs to desired values in the face of model uncertainty and disturbances.
- Feedback is the main tool to alter system behaviour. It is robust to uncertainty.
- Tradeoffs have to be made to handle constraints e.g. actuator saturation, noise amplification.

CLASSICAL CONTROL

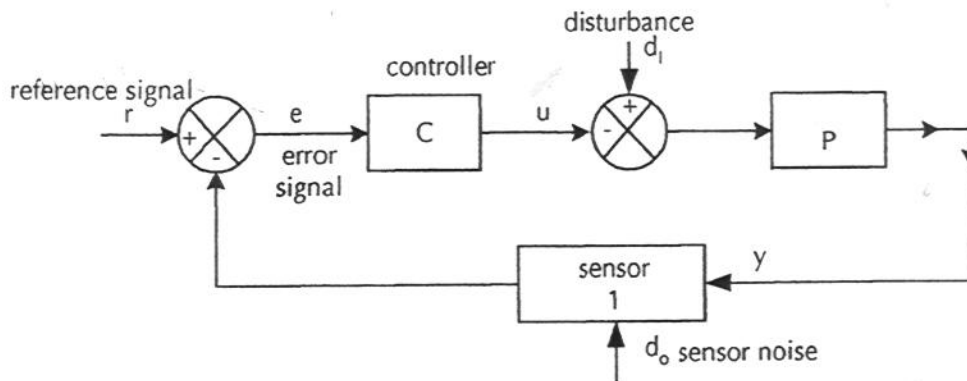


FIGURE 4.1 Classical one DOF feedback control

- Note the one degree of freedom (1 DOF) design. i.e. only one signal (here error, e) available for filtering.
- Note presence and location of disturbance or input load noise (sensor noise)
- Basic control problems:

Regulation - drive system to 0

Tracking/Servo - force system to follow reference
e.g. dc, sinusoid

CLASSICAL CONTROL Example

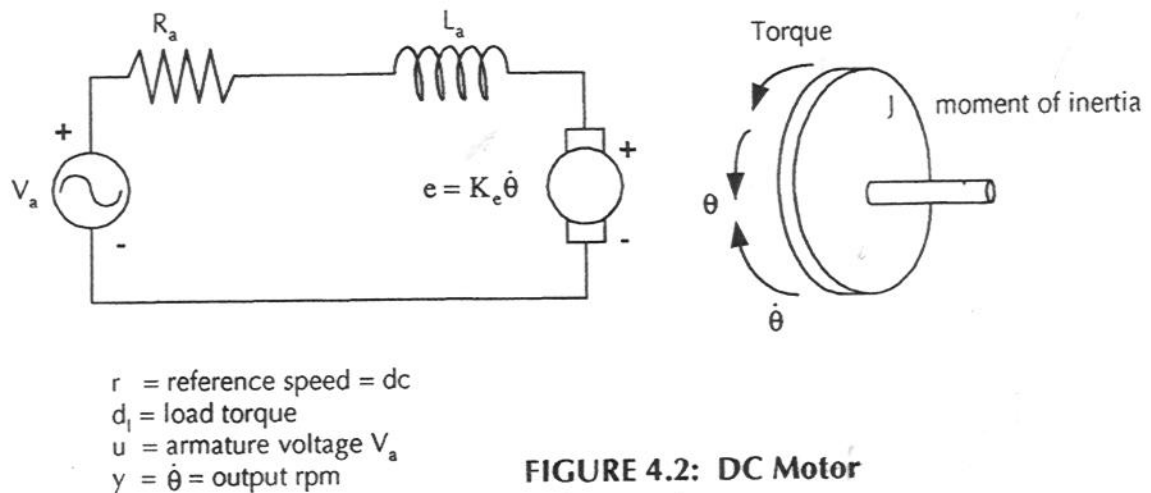


FIGURE 4.2: DC Motor

Modelling involves electrical and mechanical coupling

$$\bar{y}(s) = \frac{A\bar{u}(s) + B\bar{d}_l(s)}{(\tau_{elec}s + 1)(\tau_{mech}s + 1)}$$

Usually $\tau_{elec} \ll \tau_{mech}$ so can collapse the model to a first order system.

$$OLTF = \frac{A}{(\tau_{elec}s + 1)(\tau_{mech}s + 1)}$$

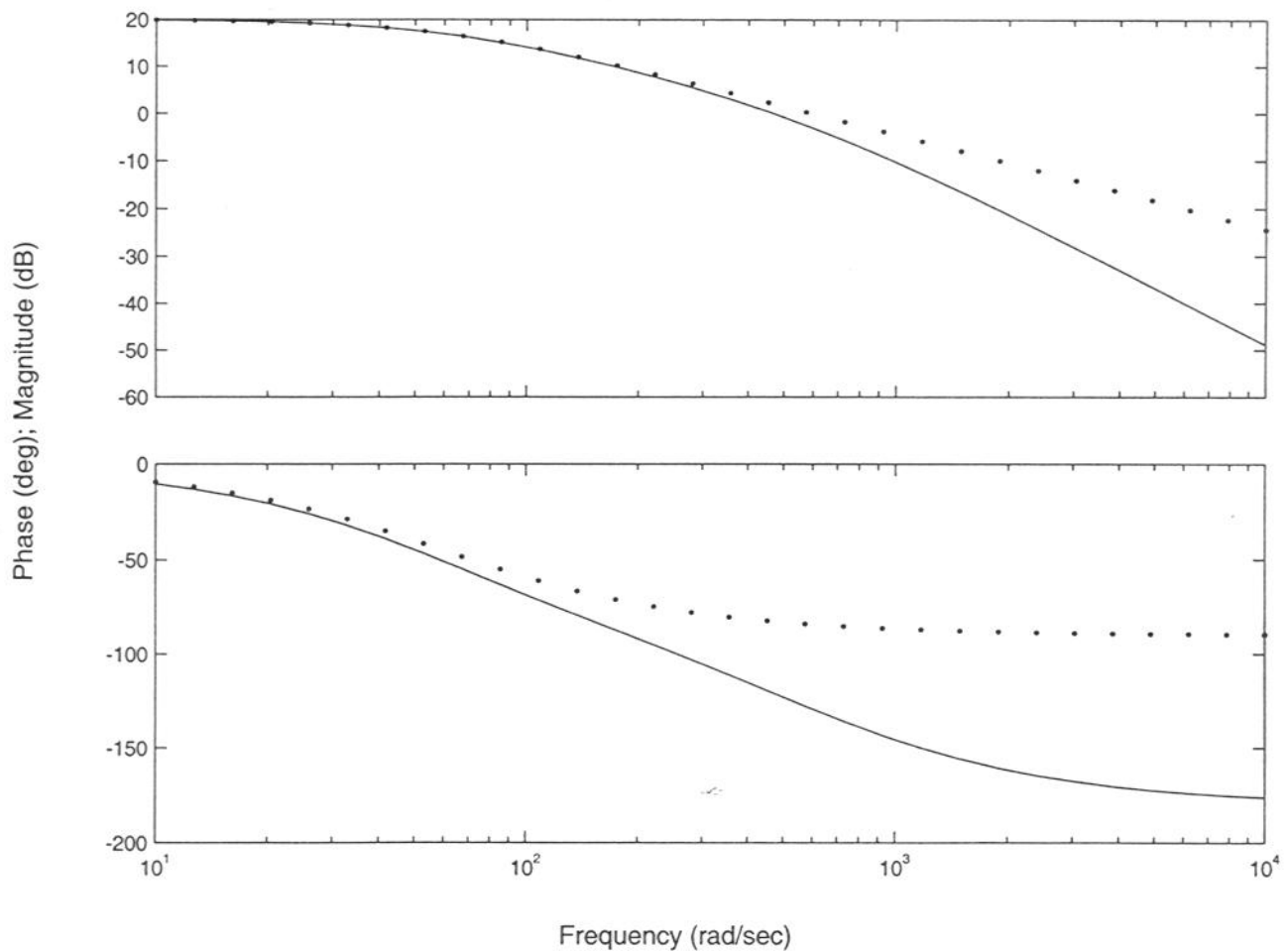
$$OL-IDTF = \frac{B}{(\tau_{elec}s + 1)(\tau_{mech}s + 1)}$$

DC MOTOR Example

$$\tau_{elec} = \frac{1}{600}, \tau_{mech} = \frac{1}{60}, A = 10, B = 50$$

Below are Bode plots of OLTF for full model and first order approximation.

DC Motor Bode Plots [Full model (-), Approx Model (.)]



CLOSED LOOP BEHAVIOUR - 1 DOF

Forward Loop

$$\bar{y}(s) = P(s)\bar{u}(s) + P(s)\bar{d}_I(s)$$

Feedback loop

$$\bar{u}(s) = C(s)[\bar{r}(s) - \bar{y}(s) - \bar{d}_0(s)]$$

Collect together (drop s argument for ease)

$$\begin{pmatrix} 1 & -P \\ C & 1 \end{pmatrix} \begin{pmatrix} \bar{y} \\ \bar{u} \end{pmatrix} = \begin{pmatrix} P\bar{d}_I \\ C\bar{r} - C\bar{d}_0 \end{pmatrix}$$

Invert

$$\begin{aligned} \begin{pmatrix} \bar{y} \\ \bar{u} \end{pmatrix} &= S \begin{pmatrix} 1 & P \\ -C & 1 \end{pmatrix} \begin{pmatrix} P\bar{d}_I \\ C\bar{r} - C\bar{d}_0 \end{pmatrix} \\ S &= S(s) \\ &= \frac{1}{1 + PC} = \frac{1}{1 + P(s)C(s)} \\ &= \text{Sensitivity function} \end{aligned}$$

Now read out solutions

Slide 7

CLOSED LOOP BEHAVIOUR - 1 DOF - Continued

$$\bar{y} = SPC\bar{r} - SPC\bar{d}_0 + SP\bar{d}_I$$

$$\bar{u} = SC\bar{r} - SC\bar{d}_0 - SPC\bar{d}_I$$

Closed loop transfer function

$$G(s) = CLTF = SPC = 1 - S(s) = T(s)$$

Input disturbance transfer function

$$IDTF = S(s)P(s)$$

Noise transfer function

$$\begin{aligned} NTF &= -SPC = -(1 - S(s)) = -T(s) \\ &= \text{Complementary sensitivity} \end{aligned}$$

SENSITIVITY FUNCTION

- Measures CL effect of model perturbations.

Let $P(s)$ change to $P(s) + dP(s)$. The relative change in CLTF is as follows:

$$\begin{aligned}
 G &= \frac{PC}{1 + PC} \\
 \Rightarrow dG &= \frac{dPC}{1 + PC} - \frac{PC}{(1 + PC)^2} C dP \\
 \Rightarrow \frac{dG}{G} &= \frac{dP}{P} - \frac{PC}{1 + PC} \frac{dP}{P} \\
 &= \frac{1}{1 + PC} \frac{dP}{P} = S \frac{dP}{P}
 \end{aligned}$$

So the smaller is $|S(jw)|$ the less the effect of model uncertainty.

SENSITIVITY FUNCTION - Continued

From previous two slides:

- For good tracking want

$$T(jw) \simeq 1 \text{ i.e. } |S(jw)| \ll 1$$

- For good disturbance rejection need

$$|S(jw)| \ll 1$$

- For good noise mollification need

$$|T(jw)| \ll 1 \text{ i.e. } |S(jw)| \text{ large.}$$

- These requirements conflict and so must be traded off against each other.

Also for a physical system $|PC(jw)| \rightarrow 0$ as $w \rightarrow \infty$, so $|S(jw)| \rightarrow 1$ as $w \rightarrow \infty$.

- So can only get $|S(jw)| \ll 1$ over limited bandwidth.

DC MOTOR CONTROL - Example - Sensitivity Functions

- Proportional Control $C(s) = K$

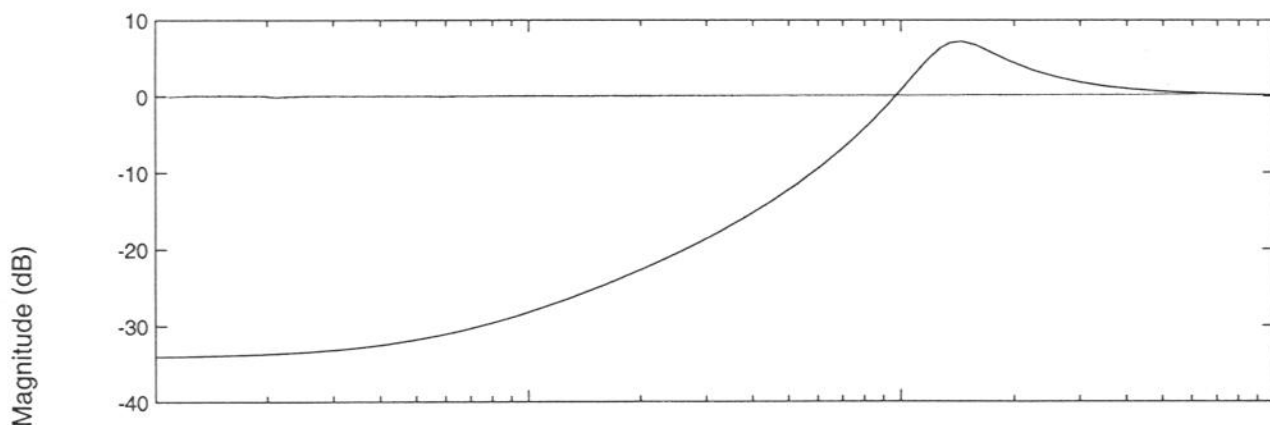
$$\Rightarrow S(s) = \frac{1}{1 + PC} = \frac{1}{1 + \frac{AK}{(1 + \tau_e s)(1 + \tau_m s)}}$$

- Proportional+ Integral Control $C(s) = K(1 + \frac{1}{T_I s})$

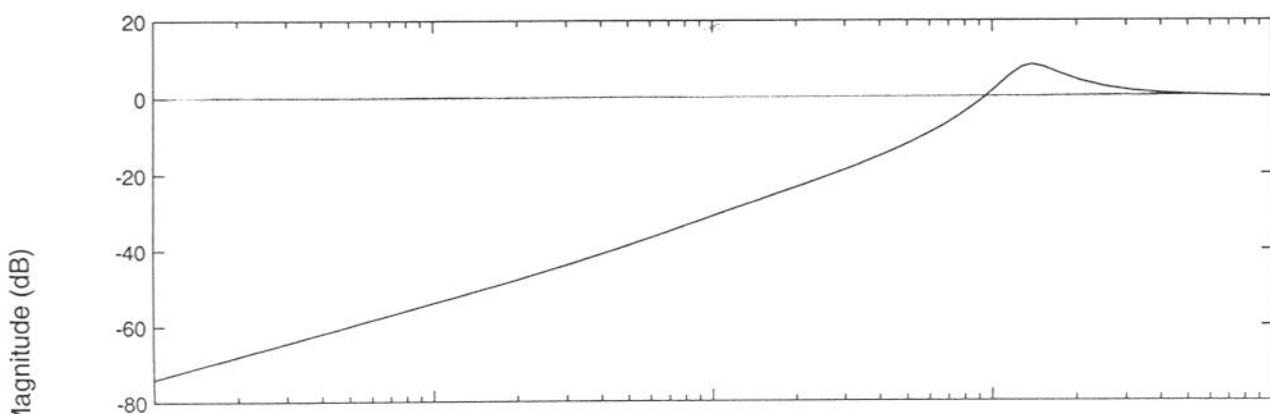
$$\Rightarrow S(s) = \frac{1}{1 + PC} = \frac{1}{1 + \frac{AK(1 + \frac{1}{T_I s})}{(1 + \tau_e s)(1 + \tau_m s)}}$$

- Example with $K = 5, T_I = .01, A = 10$

DC Motor Proportional Control Sensitivity



DC Motor Proportional+Integral Control Sensitivity



CLASSICAL CONTROL: ISSUES, TOOLS

- 1 DOF methods
 - (i) PID or lag-lead controllers.
 - (ii) Feedforward control
For disturbance/ noise cancellation when disturbance can be measured.
 - (iii) Internal Model Principle
For disturbance rejection; and for reference tracking.
 - (iv) Delay
Requires special treatment, e.g. Smith predictor.
 - (v) Tradeoffs
Due especially to RHP zeros.

CLASSICAL CONTROL: ISSUES, TOOLS - Continued

- More recent: 2 DOF methods

(i) Pole placement

By linear state feedback (LSFB)

(ii)-(v) As above.

- Multi - DOF

In general better performance is possible if more internal states are measured.

INTERNAL MODEL PRINCIPLE For Disturbance Rejection- 1 DOF Architecture

The idea is to model the disturbance

e.g. a step disturbance has

$$\dot{d}_I = 0 \Rightarrow \bar{d}_I(s) = \frac{c}{s}$$

For a sinusoid

$$d_I = a + b\sin(w_0 t) \Rightarrow \bar{d}_I(s) = \frac{c}{s(s^2 + w_0^2)}$$

In general write

$$\bar{d}_I(s) = \frac{\bar{d}_N(s)}{\bar{d}_D(s)} = \frac{\text{numerator}}{\text{denominator}}$$

$\bar{d}_D(s)$ is called the generating polynomial.

INTERNAL MODEL PRINCIPLE 1 DOF - Continued

The component of the output due to the disturbance is $SP\bar{d}_I$

$$\begin{aligned}
 &= \frac{P}{1 + PC} \bar{d}_I \\
 &= \frac{\frac{b}{a}}{1 + \frac{b}{a} \frac{c_N}{c_D}} \frac{\bar{d}_N}{\bar{d}_D} \\
 &= \frac{bc_D}{ac_D + bc_N} \frac{d_N}{d_D}
 \end{aligned}$$

Now $ac_D + bc_N$ is the denominator polynomial of the CLTF and so must be stable.

So we need $c_D(s)$ to have $d_D(s)$ as a factor to get overall stability [the roots of $d_D(s)$ be on the imaginary axis]. This is the:

- IMP:denominator of $C(s)$ must have $d_D(s)$ contained within it for disturbance rejection.

IMP - Example - DC MOTOR

- Simplified Motor Model $P(s) = \frac{1}{1+\tau_m s}$

Step Disturbance LT : $\bar{d}_I(s) = \frac{c}{s} = \frac{\bar{d}_N(s)}{\bar{d}_D(s)}$

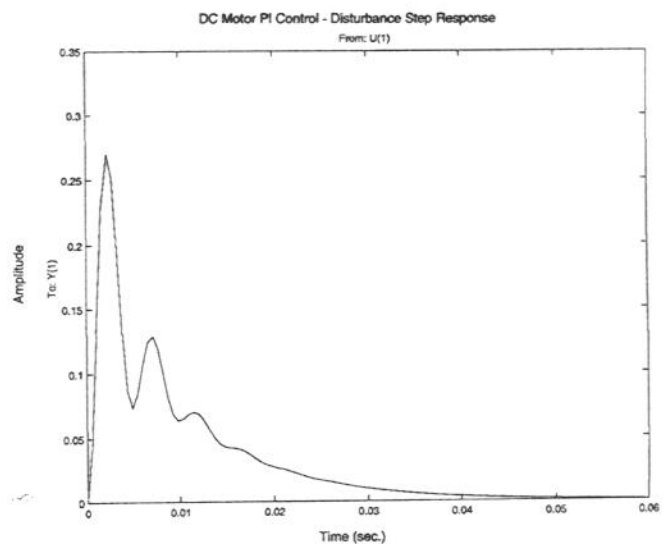
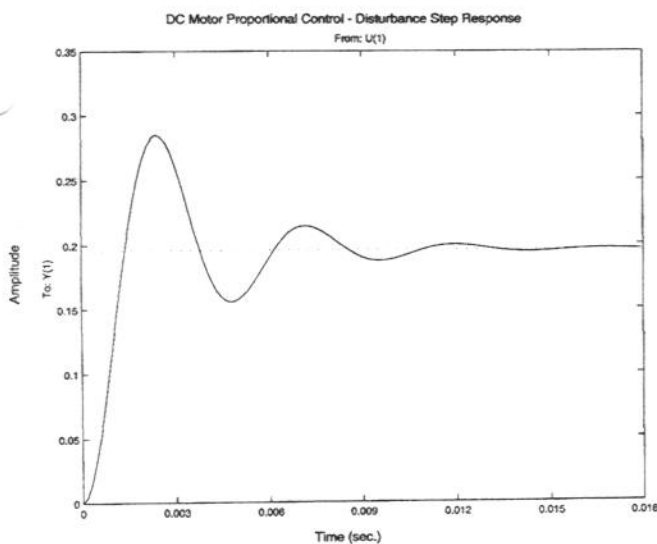
So IMP requires s is a factor of $c_D(s)$

- Proportional Control $C(s) = \frac{K}{1}$

thus cannot reject a step disturbance.

- P+I Control $C(s) = K(1 + \frac{1}{T_I s}) = \frac{K(1+T_I s)}{T_I s}$.

So PI can reject a step disturbance.



TWO DOF

- The idea is that 2 signals are available for filtering / manipulation. There are many equivalent architectures. We give perhaps the 3 major ones.

- Inverse Model

Suited to stable plants

Includes a nominal plant model

The two filtered signals are a , e

- LSFB + observer

Suited to pole placement

Filtered signals are u, y

- Reference Signal Prefiltering

Useful when reference is time varying

Filtered signals are r, a

2 DOF architectures

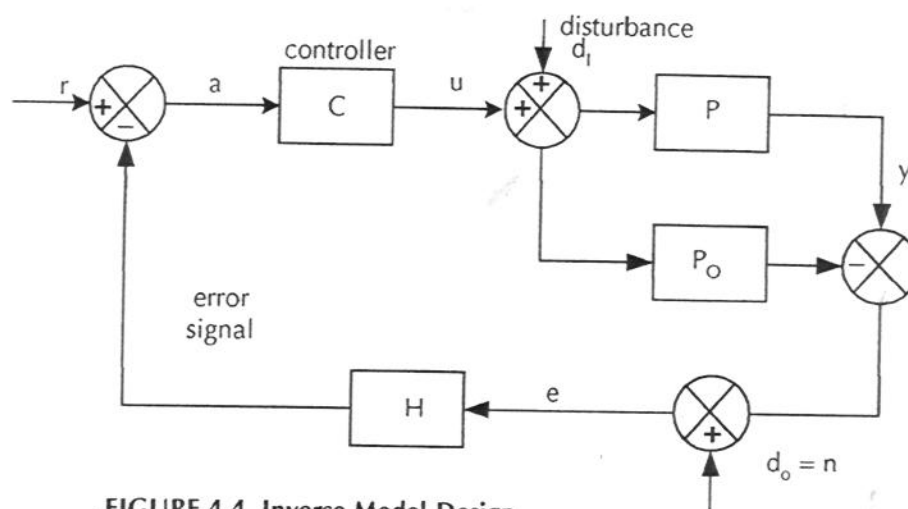


FIGURE 4.4 Inverse Model Design

P_o = nominal model TF
 a = actuating signal

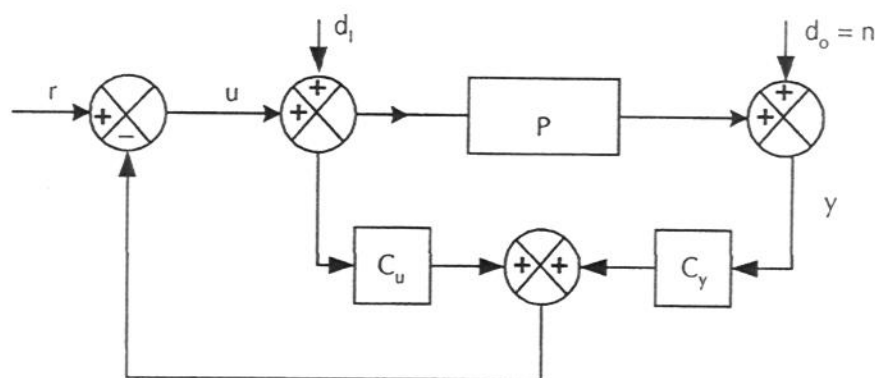


FIGURE 4.5 LS FB + Observer

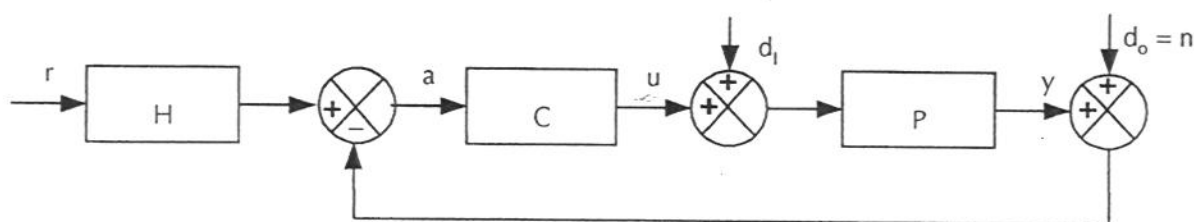


FIGURE 4.6: Prefiltering of Reference Signal

2 DOF Block Diagram Algebra

Consider the prefiltering architecture.

Forward Loop

$$\bar{y} = P\bar{u} + P\bar{d}_I$$

Feedback Loop

$$\bar{u} = C[H\bar{r} - \bar{y} - \bar{d}_0]$$

Collect together

$$\begin{pmatrix} 1 & -P \\ C & 1 \end{pmatrix} \begin{pmatrix} \bar{y} \\ \bar{u} \end{pmatrix} = \begin{pmatrix} P\bar{d}_I \\ CH\bar{r} - C\bar{d}_0 \end{pmatrix}$$

Solve

$$\begin{pmatrix} \bar{y} \\ \bar{u} \end{pmatrix} = S \begin{pmatrix} 1 & P \\ -C & 1 \end{pmatrix} \begin{pmatrix} P\bar{d}_I \\ CH\bar{r} - C\bar{d}_0 \end{pmatrix}$$

$$S = \frac{1}{1 + PC}$$

Now read out individual closed loops

2 DOF BLOCK DIAGRAM ALGEBRA - Continued

$$\begin{aligned}\Rightarrow \bar{y} &= SPCH\bar{r} - SPC\bar{d}_0 + SP\bar{d}_I \\ \bar{u} &= SCH\bar{r} - SC\bar{d}_0 - SPC\bar{d}_I\end{aligned}$$

And as before we can read important TFs

$$CLTF = SPCH = H(I - S) = HT = H \frac{PC}{1 + PC}$$

$$IDTF = SP = \frac{P}{1 + PC}$$

$$NTF = -SPC = -(1 - S) = -T$$

We can now manipulate $CLTF$ and NTF independently

C determines NTF

(C and) H determines $CLTF$